

Enabling Data Collection and Analysis for Precision Agriculture in Smart Farms

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Abstract—This article presents an in-depth exploration of multifaceted efforts in agricultural research aimed at addressing the unpredictable nature of crop production and related processes, including the demonstration of data collection and its application. This research focuses on leveraging current technologies and devising sustainable solutions to mitigate uncertainties attributed to natural climatic conditions and infectious agents. The central theme of this review centers around the utilization of Internet of things sensors for data collection, cloud software for data processing, and the integration of diverse machine learning algorithms for data analysis. The objective is to advance insights into the application of these technologies in agriculture and their potential to enhance crop yield and sustainability. The article comprehensively explores the technological landscape and the levels at which current research is being conducted, shedding light on machine learning algorithms, their specific functions, and comparative analysis of each algorithm based on different use cases. Furthermore, the article presents an implementation approach that integrates sensors and machine learning. Its primary application is to predict the type of crop to produce based on nutrient levels detected by the sensors.

Index Terms—Cloud processing, Internet of Things (IoT), machine learning (ML), precision agriculture, wireless sensor network (WSN).

NOMENCLATURE

Abbreviations

ML	Machine learning.
RF	RandomForest.
DL	Deep learning.
AI	Artificial intelligence.
ANN	Artificial neural network.
CNN	Convolutional neural network.
RNN	Recurrent neural network.
GAN	Generative adversarial network.
GRU	Gated recurrent unit.
CatBoost	Categorical boosting.

Received 5 February 2024; revised 19 April 2024, 24 June 2024, and 2 September 2024; accepted 2 September 2024. Date of publication 2 October 2024; date of current version 11 April 2025. This article was recommended by Editor-in-Chief D. Demarchi. (Corresponding author: Fru Ngwa Fru Junior.)

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Digital Object Identifier 10.1109/TAFE.2024.3454644

LightGBM	Light gradient boosting machine.
XGBoost	Extreme gradient boosting.
MAPE	Mean absolute percentage error.
NDVI	Normalized difference vegetation index.
SAVI	Soil adjusted vegetation index.
PWM	Pulsewidth modulation.
IoT	Internet of Things.
LoRaWAN	Long-range wide area network.
EMP-ZBR	Energy-aware multipath zone-based routing.
WSN	Wireless sensor network.
Wi-Fi	Wireless fidelity.

I. INTRODUCTION

AS A result of the unpredictable nature of crop production and various agricultural processes, primarily attributed to natural climatic conditions and infectious diseases, a significant amount of research has been carried out. The aim of this research is to address these uncertainties by reviewing the latest work done using current technologies and devising sustainable solutions. This review will primarily explore the use of IoT sensors to collect data, the application of cloud software for processing, and the implementation of several ML algorithms in regard to the data. In addition, we discuss other approaches used in different research works that aim to produce solutions for similar problems.

Many economies are sustained by their various agrarian sectors; hence, it is crucial to determine the best course of action to mitigate any decrease in crop yield. For instance, in the case of water management, sensors are used to address these challenges in agriculture through the precise monitoring of soil moisture levels. The integration of wireless sensor technology and precision agriculture practices is presented as a promising approach for optimizing agricultural decision-making and improving crop performance [1]. Presently, there is a considerable amount of research being conducted on agricultural IoT technology, with a broad scope and in-depth focus. However, the practical implementation of this technology in agriculture is primarily in an experimental demonstration phase.

Apart from the different technologies associated with the extensive amount of research done, this article delves into the various levels at which current research has been conducted. In addition, the ML algorithms utilized and their intended functions are investigated. The diverse approaches to implementation are explored, including a proposed methodology of precision agriculture with IOT sensors to sense different soil and weather

parameters simultaneously and use the data to build an RF model to recommend most suitable crops to grow. Finally, common challenges encountered in various papers are addressed, research gaps are identified, and future developments with a similar objective are proposed.

This article is organized as follows.

In Section II, we discuss the data collection process using sensors and how various network protocols are used for data transfer, depending on the scope of precision agriculture.

In Section III, we investigate various ML algorithms, ranging from less complex to more complex algorithms, that are used for AI to find patterns and trends and to replicate basic functionalities used in precision agriculture.

In Section IV, we propose our own methodology for precision agriculture by building an RF model to recommend the best crop to grow based on agricultural parameters.

Section V discusses real-life applications of precision agriculture with other technologies apart from IoT sensors.

In Section VI, we look at challenges, regulatory policies, and research gaps concerning precision agriculture.

Finally, Section VII concludes the article by highlighting all the findings discussed in the previous sections.

II. IOT SENSORS IN AGRICULTURE

IoT sensors are primarily used for communication between multiple devices over a network. In these cases, the data have a short lifespan, fulfilling their immediate purpose and then becoming obsolete. However, in this article, we explore scenarios where IoT sensors are used solely for long-term data collection with extended goals. In these situations, the devices do not need to communicate with each other or other devices. The IoT sensors are repurpose to sense data without expecting incoming data, simplifying the configuration and setup.

In the agricultural sector, many researchers have employed similar methodologies for precision agriculture. For instance, Puengsungwan [2] implemented a real-time soil moisture monitoring IoT-based system where the data are sent to the cloud. The entire system is powered by solar panels, making it more autonomous.

In addition, Hong et al. [3] explored a data-driven technique for predicting soil moisture days in advance by using two soil moisture sensors to sense moisture at depths of 5 and 10 cm, with each data collection interval set at 40 s across nine different sites. Another sensor used is the temperature and humidity sensor for data collection.

In [4], the authors tested a DS18B20 soil temperature sensor to provide a live feed of soil temperature to farmers, achieving a 99% accuracy rate in sensing and providing real-time temperature analysis. Most papers employ sensor fusion with microcontrollers to collect data on multiple agricultural parameters simultaneously.

In [5], all sensors are connected to an ESP32 Node MCU, which provides Wi-Fi and Bluetooth connectivity with higher computational power compared to the ESP8266, which only offers Wi-Fi connectivity and is used for simpler tasks. The work done in [5] utilized pre-existing cloud software as a service (Blynk). In [6] and [7], they simulate data collection using

edge computing, reducing latency, lowering power consumption, improving resource management, enabling scalability, and enhancing data security.

Kumar [8] used an Arduino Uno board with the available Arduino IDE to connect soil moisture, raindrops, and DHT11 sensors to collect data and visualize it on a serial LCD display. This is a very simple setup, but a Wi-Fi module was included in the setup. Conversely, Raviteja and Supriya [9] implemented a monitoring system with Wi-Fi connectivity using the ESP8266, which is smaller and ideal for fast real-time system monitoring in precision agriculture. The ESP32 is suited for functions with rapid input and output processes, as Venu [10] used the ESP32 to turn a water pump ON or OFF based on the soil moisture level. All these instructions are programmed into the microcontroller using the Arduino IDE. On the other hand, the Raspberry Pi is a standalone hardware platform with an operating system for data collection and processing and has built-in capacity for more complicated wireless communication (see Table I).

A. Networks for Data Transmission

Reliable data transmission is crucial for collecting information from sensors, and the most common and preferred method for this is to explore wireless communication. Various transmission technologies are employed depending on the specific task at hand. This section explores the work of different research papers that are specific for large and fast data transmission over wide area.

1) *Long Range (LoRa)*: Long Range, or LoRa, is a wireless communication technology that has a reputation for efficiency and increased reach. Utilizing low-power, wide-area networks, it is perfect for extended connectivity while consuming the least amount of energy. Because of its design, LoRa can transmit data over long distances and is, therefore, appropriate in situations where extensive cabling is not practical. Due to its dependability in demanding conditions and low signal-to-noise ratios, which are typical in remote locations, it is the best option for applications that prioritize data accuracy, such as sensor networks.

The use of LoRa in this study demonstrates a calculated strategy for reliable, long-distance data transmission that protects data integrity. This section explores papers that use this network technology for data transmission (see Table II).

2) *Wireless Fidelity*: Wi-Fi, which is well known for being adaptable and widely used, is now essential for enabling smooth connectivity for a wide range of applications. Wi-Fi, which uses radio frequency signals to function, allows devices to transfer data quickly, which makes it a great option in situations where reliable and effective communication is needed. Among the networks examined in this article, Wi-Fi stands out as being more reliable for outdoor data transmission, even with various obstacles, and it also supports the transmission of multimedia data, such as images. This section examines studies that use Wi-Fi technology for data transmission, providing insight into how it can be used to guarantee stable connectivity for sensor data collection (see Table III).

3) *Zigbee*: A wireless communication technology that is well known for its adaptability and effectiveness has become

TABLE I
MOST COMMON MICROCONTROLLERS WITH FUNCTIONAL FEATURES

Microcontroller	Weight	RAM	General Purpose Input/Output (GPIO) pins	WI-FI	Bluetooth	Bandwidth for data transfer	Price Range(\$)
Arduino uno	25 g	2 kb SRAM	14 (6 PWM)	×	×	115.2 kb/s	20–30
ESP8266	1.5 g	64 kb SRAM	17	✓	×	72 Mb/s	3–10
ESP32	7 g	520 kb SRAM	34	✓	✓	150 Mb/s	6–12
Raspberry Pi 4	46 g	1 gb, 2 gb, 4 gb & 8 gb	26	✓	✓	433 Mb/s	35–75

TABLE II
STUDIES UTILIZING LoRA TECHNOLOGY

Ref	Year	Functionality	Key features	Network Setup	Data Transmission
[11]	2019	Building a mobile gateway for agriculture with LoRaWAN support.	Offers a solution that combines adaptability, reach, and the capacity to cover wide areas.	DHT11 sensor connected to LoRaWAN sensor node to collect and transmit data to mobile LoRaWAN gateway	Mobile LoRaWAN gateway operating at 915 MHz with a dedicated omnidirectional fiberglass antenna providing at least 10 dBi gain.
[12]	2022	Use sensors to detect field conditions in real time, and the LoRa Gateway is used to receive and send the data via Ethernet or Wi-Fi to the cloud server.	The application of WSN and LoRa technology to improve connectivity, data collecting, and automation in IoT agricultural activities.	Sensors are directly connected to the LoRaWAN gateway which acts as receiver and transmitter of data	long-range (up to 10 km), low-power (66 mW per hour) communication platform that senses and transmits data every 30 s.
[13]	2020	Utilizing LoRaWAN technology to maximize battery life for post-harvest storage in agriculture	Post-harvest agriculture using LoRaWAN technology, a thorough assessment of battery life, optimization suggestions, and a thorough investigation of hardware components	The LoRa network is composed of four key components: a sensor node, a LoRa gateway, a network server, and an application server.	A Raspberry Pi 3 is employed to forward the packets to the Things Network server using 802.11n WiFi on the 2.4 GHz band
[14]	2021	Wide Area soil monitoring system of agricultural Parameters	Central LoRa gateway node uploads data to the cloud server every 15 s	Three LoRa environmental nodes placed at distances 350 m, 400 m, and 600 m, respectively, transmit data to Single LoRa gateway	LoRa gateway utilizes a separate unlicensed 2.4 GHz Wi-Fi frequency band to connect with a cloud storage server.

a major force in the data transmission industry. Zigbee, which is intended for low-power, short-range applications, offers a dependable way to link sensors and devices in a range of environments. Zigbee networks, which run on the IEEE 802.15.4 standard, perform best in situations where power consumption is a crucial factor. Zigbee's mesh networking features make it more resilient and allow for smooth communication even in difficult settings.

Zigbee is a compelling option for some tasks because it can be used for applications that require low data rates and periodic sensor data transmission. This section examines how Zigbee technology is used in research papers and clarifies how it helps ensure dependable data transmission for particular use cases (see Table IV).

B. Cloud Technology

With recent advancements in technology, it is now possible to transmit data to a remote server while maintaining access to that data. This technology is commonly referred to as cloud technology.

The cloud computing layer is one of the four integral components in the system, alongside the agricultural data collection layer, edge computing layer, and data transmission layer [23]. In the context of agriculture IoT sensor monitoring networks, cloud computing serves as the essential component responsible for processing the data collected from IoT sensors [24]. Some research also incorporates custom cloud providers, which are IoT platforms for analytics services designed to store, visualize, and analyze data [25]. This cloud infrastructure primarily facilitates the visualization and monitoring of nodes within the system, further expanding its utility. Moreover, the data are utilized through the application of ML algorithms to explore the additional use cases of the collected data. Cloud-based technology empowers farmers to store and share vital information related to crops, such as fertilizer prices and crop prices, in a cloud-based database. Furthermore, it enables the application of Big Data predictive analytics techniques for analyzing and forecasting crop yields and fertilizer costs. The adoption of cloud-based systems also paves the way for the development of efficient and well-defined applications [26]. The paper [27] explores the successful deployment of precision technology in Ireland. Through surveys and interviews

TABLE III
STUDIES UTILIZING WI-FI TECHNOLOGY

Ref	Year	Functionality	Key features	Network Setup	Data Transmission
[15]	2021	Using Wi-Fi to establish a dense network of nodes equipped with diverse sensors to gather data from the soil, plants, and atmosphere.	The data collected is used to compute the plant's irrigation requirements(ET _c) and synchronize with the irrigation system for timed irrigation.	Virtual Private Network (VPN) server is configured on a router to facilitate network maintenance and provide access to data logger and the fertigation controller.	A 19 dBi 120° sector antenna that facilitates a 5 GHz (802.11ac) link. The link ensures low-latency, high-throughput data transfer between the sensor nodes, data loggers, and remote management systems.
[16]	2022	Finding impact on signal coverage and the number of Access Points needed for effective operation in a WiFi-based WSN.	Designing a WSN to estimate the number of sensors and access points needed to cover a field with different areas for different conditions.	Cluster-based topology of sensors and with router responsible for managing communication within the cluster and relaying Access Point.	The best coverage for the sensor nodes is achieved at a lower height, specifically at 0.5 meters.
[17]	2019	Examining and verifying the usefulness of using commercial WiFi systems to track movements on a farm.	Evaluate the capability of WiFi systems to deliver positional information, even in environments with restricted access.	Two ESP8266 connected to Wi-Fi are used to send positional coordinates.	Cisco Meraki manages access points, which serve as central nodes in the wireless network.
[18]	2020	Calculating soil moisture content and generates distribution maps to manage irrigation effectively.	RaspberryPi serves as Base Station Unit(BSU) to store and transmit sensor node data to the cloud.	BSU acts as a central hub, connected to multiple Terminal Units (TUs) equipped with sensors for monitoring soil moisture, temperature, and humidity	The BSU sends a command to all TUs to publish their data to specific MQTT topics. Each TU publishes its data, and the BSU subscribes to those topics to receive the data via the MQTT broker.
[19]	2018	Establishing an affordable, effective, and deployable WSN for use in monitoring applications.	Integration of ESP8266 modules with smart-config technology, the removal of gateway nodes, efficient neighbor discovery in WSN for monitoring applications.	Sensor nodes equipped with ESP8266 Wi-Fi modules connect directly to the Wi-Fi access point using Smart-Config technology	The number of sensor nodes varied from 1 to 5 in a 250m x 250m area. UDP provides better E2E delay, reducing it by 1.3 s for ESP8266

TABLE IV
STUDIES UTILIZING ZIGBEE TECHNOLOGY

Ref	Year	Functionality	Key features	Network Setup	Data Transmission
[20]	2024	Investigating the impact of outdoor data collection of zigbee network on the field.	Operates within 2.4–2.5 GHz, supports 16 channels, But does not natively support image formats (JPEG, PNG).	Node MCU ESP8266 controllers act as central hubs, connected to ZigBee devices at both the receiver and sender sides. A sensor unit, linked to the sender side Node MCU via Wi-Fi, collects data. The ZigBee transmitter on the sender side communicates with the receiver.	Data from the sensor unit was sent over Wi-Fi to the ZigBee device on the sender side, which was then forwarded to the ZigBee receiver through the ZigBee connection.
[21]	2023	Greenhouse monitoring system using WSNs via Zigbee connectivity and sensor nodes.	EMP-ZBR routing protocol is designed to reduce end-to-end delay in data transmission.	Zigbee -based Wireless Mesh Network topology features coordinators, routers, and terminal nodes arranged in layers, allowing routers to communicate directly with one another.	EMP-ZBR ensures efficient, reliable data transmission by reducing delays and boosting packet delivery through optimized path selection, minimizing congestion in agricultural sensor networks.
[22]	2021	Monitoring system for agricultural fields that utilizes ZigBee wireless modules to collect and transmit data on soil moisture, temperature, atmospheric pressure, and sunlight intensity.	The system is designed to collect data in real time through Zigbee device connections.	The network setup consists of a star topology where ZigBee end devices collect environmental data and transmit it to a central ZigBee Coordinator, which then sends the aggregated data to a data center for analysis and monitoring.	Data transmission is based on the CC2530 ZigBee SoC, which is suitable for low-power WSNs.

with 32 farmers, this study delves into the factors influencing technology adoption in the agricultural sector in Ireland. Over 60% of farmers in the region have embraced technology, with 70% incorporating cloud technology into their agricultural practices.

Notably, individuals between 25 and 44 years old exhibit a higher adoption rate, especially in dairy farming. The utilization of technology varies across farming sectors; dairy farmers employ decision support systems and remote monitoring tools, while beef and sheep farmers lean toward diet feeder technology

TABLE V
USE CASE ANALYSIS OF MACHINE LEARNING ALGORITHM IN AGRICULTURE

Algorithm	Analysis
Multiple Linear Regression	Multiple linear regression is employed to assess the impact of various factors such as fertilizer and water on crop yields, establish relationships between explanatory variables and crop yield, and predict annual yield through parameters, such as origin, type, land, area, and season within a model [33], [34].
Decision tree	Decision Trees are utilized to identify predictor variables that significantly impact agricultural production and are used in intelligent irrigation systems and crop recommendation systems within the agriculture industry [35], [36], [37].
Random Forest	RF algorithm is used in predicting crop yield and crop recommendation system [38], [39], [40], [41].
SVM	SVM is used in an intelligent irrigation system for the agriculture industry, performing tasks such as crop prediction and classification of agricultural data [42], [43].
Neural Network	Neural networks (ANN and CNN) are utilized in agricultural systems for predicting crop yield and determining suitable crops and fertilizers based on geographic location and soil quality, as well as addressing real-life challenges such as crop and land cover classification, detection of harvested field status, and verification of pests and diseases [44]. Inputs to a Neural Network can be both textual and visual imagery.
KNN	KNN are utilized both in a crop recommendation system and for mapping land cover and land use with a limited number of discrete classes [45].

and weather apps. There is a growing interest in leveraging cloud technology for tasks such as animal tracking, online applications, and yield mapping.

Despite this enthusiasm, barriers to adoption persist, including the perceived complexity of available cloud-based applications. Farmers express a need for more user-friendly and mobile-compatible solutions, particularly for herd management and financial budgeting. This presents a potential market opportunity for Agro-Tech companies to develop tailored solutions to meet farmers' needs.

1) *Economic and Social Implications of SensorCloud Adoption in Farming*: SensorCloud technologies offer significant potential to enhance various aspects of agriculture, including production, efficiency, and sustainability. By providing real-time data, these technologies enable farmers to make informed decisions, ultimately leading to increased agricultural productivity and income. However, the widespread adoption of SensorCloud technologies also brings potential adverse impacts, such as the intensification of agriculture and the depletion of natural resources. Despite these challenges, the overall benefits of utilizing SensorCloud technologies in agriculture are evident, with the potential to revolutionize farming practices and mitigate some of the challenges facing the industry [28], [29].

Examining the cost benefits of SensorCloud technology sheds light on significant advantages for both farmers and end-users. Farmers experience a notable 33.83% reduction in cash outflow compared to traditional WSNs, while end-users enjoy a 14.72% decrease in expenditure. Cloud service providers (CSPs) play a crucial role in managing cash flows, with their profits influenced by rental fees and end-user payments. The integration of SensorCloud technology demonstrates cost-effectiveness through

a substantial decrease in energy consumption by 36.68%, underscoring the importance of efficient resource allocation to ensure optimal performance and profitability for CSP [30].

In addition, contrasting how sensor cloud technology is utilized in small-scale farming versus large-scale farming contexts yields valuable insights into its potential effects. Smallholder farmers confront distinct challenges related to resource availability, decision-making, and cost-effectiveness, necessitating customized solutions for fair access and efficient utilization of sensor data.

In contrast, large-scale farms may employ sensor cloud technology for more advanced decision-making and resource management strategies.

III. DATA UTILIZATION METHODS AND FUNCTIONS

A. Machine Learning Algorithm Applied

Using ML algorithms on this data is the goal of many papers. ML creates models for particular applications using training data and verifies the model's effectiveness in real-world scenarios using test data. Developing ML models improves flexibility and reusability, particularly for tasks such as large-scale dataset classification and prediction (see Table V).

The K-nearest neighbor (KNN) method can recommend crops based on their similarities in the context of crop recommendation systems, just as it can with environmental factors such as soil profile, temperature, and meteorological data [31]. Additional studies in agriculture data and ML investigate hybrid ML, in which a number of straightforward algorithms work together to solve problems that would be impossible for them to solve

separately. This strategy can be modified for use with different crops and environmental circumstances [32].

B. Functionality of Machine Learning Algorithms

1) *K-Mean*: In precision agriculture, K-Means clustering transforms intricate agricultural data into meaningful clusters, enabling targeted and efficient resource management. For example, in various scenarios with limited data points (image), K-Means segments fields into distinct zones based on soil composition, moisture levels, and crop health. This segmentation facilitates tailored resource application, leading to optimized yields.

2) *Support Vector Machine (SVM)*: SVMs are very good at handling nonlinear relationships, which makes them ideal for binary classification tasks. SVMs are especially useful in situations where the datasets are small to medium sized. Also, their sensitivity to noise levels determines how well they perform in tasks with moderate to low noise levels.

3) *Decision Tree*: Decision Trees are simple classification and regression models that can excel in tasks requiring less data with clear decision boundaries and less complex interactions. This means tasks, such as weather prediction and complex yield prediction, are not well-suited for Decision Trees. However, they are effective for tasks such as disease detection and the classification of plants and fruits.

4) *Random Forest*: RF is a type of ensemble ML algorithm. It combines the prediction of multiple models to produce a more accurate and robust result. In the agricultural sector, we deal with more complex parameters that normal classification or regression-based ML algorithms will not perform well with. RF is both a classifier and regression model, making it highly versatile for various tasks, such as predicting crop yields, classifying types of crops, detecting diseases, and even analyzing soil properties, thereby significantly improving decision-making and efficiency in agriculture.

5) *Gradient Boosting*: The Gradient Boosting algorithm operates by constructing models sequentially, where each new model aims to rectify the mistakes of its predecessor. It is an ensemble ML technique utilized for both regression and classification tasks. This iterative strategy frequently results in enhanced predictive accuracy, a vital aspect for predicting crop yields and detecting diseases. Nevertheless, Gradient Boosting can incur substantial computational costs and may suffer from overfitting if not meticulously fine-tuned.

The XGBoost algorithm is a tree-based method utilized for classification, regression, and ranking tasks, supporting custom loss functions. It optimizes learning by assigning higher weights to leaves with greater loss, prioritizing those with higher errors. Its efficient memory usage contributes to faster computations without sacrificing accuracy.

CatBoost is particularly suited for predicting categorical features. It employs a sophisticated ensemble learning technique that relies on binary decision trees as base predictors. During the training process, CatBoost constructs multiple decision trees sequentially, each aimed at minimizing the loss function. This iterative approach results in a highly optimized model for categorical feature prediction.

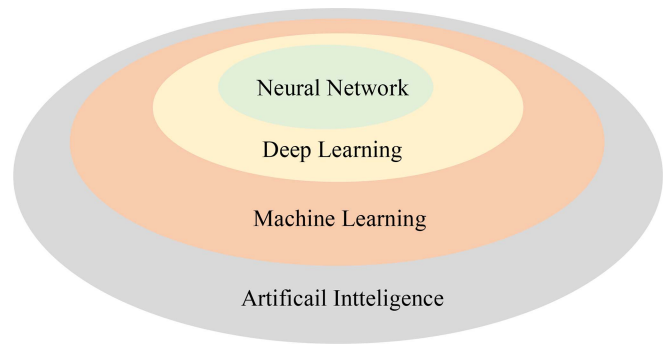


Fig. 1. Relation between DL, ML, and AI.

LightGBM novel techniques called gradient-based one-side sampling (GOSS) and exclusive feature bundling (EFB) are used to improve both efficiency and accuracy. GOSS focuses on the most informative instances by retaining those with the highest gradients, while EFB bundles mutually exclusive features to reduce dimensionality. These innovations allow LightGBM to handle large-scale data with high speed and low memory consumption, making it an excellent choice for a wide range of ML tasks.

6) *Deep Learning*: ANNs prove to be highly effective in addressing intricate problems characterized by extensive datasets, notably excelling in tasks such as image and speech recognition. Their remarkable capability lies in their adeptness at capturing complex patterns and discerning nonlinear relationships within the expansive datasets. Despite this prowess, there are instances where the application of ANN may encounter limitations. For instance, their utility tends to diminish in scenarios characterized by relatively small datasets, where the emphasis on interpretability becomes pivotal. Moreover, in situations where computational resources are constrained, the efficacy of ANN may be compromised.

ANNs serve as generalized neural network architecture. However, other specialized neural network algorithms, such as CNNs, RNNs, and GANs, have been developed to address specific challenges and optimize performance across a diverse array of tasks. These algorithms leverage distinct design principles; for instance, CNNs exploit spatial hierarchies for image recognition, RNNs utilize sequential dependencies for time series analysis, and GANs employ adversarial training for generating realistic synthetic data thereby harnessing their unique strengths to achieve superior results in their respective domains.

From Fig. 1, DL, as a part of ML and both being subsets of AI, provides the foundation algorithms that enable precision agriculture. These algorithms enhance the efficiency, productivity, and sustainability of agricultural practices by leveraging data-driven insights and automating complex tasks.

C. Comparative Analysis of Machine Learning Algorithm

These ML algorithms in Section III-B represent the latest and most commonly utilized techniques in modern precision agriculture applications. The section aims to provide a basic overview of the functionalities of ML algorithms to facilitate their effective integration into the agricultural sector. The algorithms

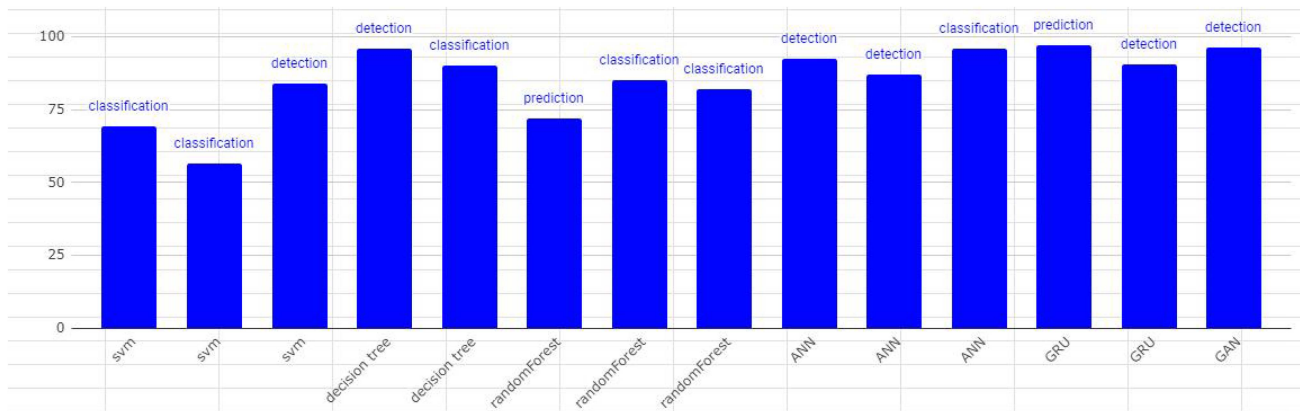


Fig. 2. Graph showing accuracy and purpose of ML algorithms that used data type “Image.”

are categorized, and analyses based on their data types are done. As the diversity and intricacy of ML algorithms continue to expand, comprehending their relative strengths and weaknesses becomes paramount for selecting the appropriate algorithm for specific tasks.

To begin with, the algorithms presented in Section III-B can be broadly categorized into supervised and unsupervised ML. Supervised ML learns from provided data points to perform predefined tasks, whereas unsupervised ML algorithms do not require predetermined data and operate solely on the input data. Section III-C1 delves into the k-means clustering algorithm, which is an unsupervised learning technique. Despite yielding favorable outcomes, these algorithms tend to be ineffective when distinguishing subtle differences in features.

The subsequent sections focus on supervised ML algorithms, which excel in capturing intricate patterns that may be challenging to discern. These algorithms demonstrate proficiency in handling numerical and image data by iteratively learning subtle patterns within the dataset.

Fig. 2 illustrates the algorithms discussed in Section III-B, which were employed for constructing ML models tailored for precision agriculture using image data. Among these algorithms, the SVM exhibits lower classification accuracy compared to the decision tree, RF, and ANN, with the latter demonstrating the highest accuracy. ANN proves to be a superior choice for image classification tasks due to its capability of comprehending intricate patterns, including edges and textures. Notably, CNN stands out as a prominent algorithm capable of learning low-level features, such as edges and textures in initial layers, gradually advancing to higher level features such as shapes and objects in deeper layers.

For anomaly detection or pattern-based tasks, ANN remains the preferred ML algorithm. In [54], the high accuracy observed can be attributed to the classification of only two features and the availability of a robust dataset. GAN utilizes neural networks in scenarios where data scarcity impedes training, although this may result in underfitting, where the model fails to capture the complexity of the data patterns. GAN models are better suited for when dealing with image data type.

From the findings in Fig. 2, GRU, a variant of RNN, emerges as the optimal choice for predicting continuous numeric data

types such as time series. In a specific scenario, the image was transformed into a time series using CNN for prediction purposes. Even in predictive tasks such as crop yield estimation, RNN surpasses RF. Nevertheless, RF stands out among other algorithms for its versatility in prediction, classification, and detection tasks. Its ease of implementation and ensemble learning nature, which harnesses multiple decision trees to yield accurate outcomes, contribute to its prominence. Consequently, the marginal disparity in accuracy between ANN algorithms is not a significant concern (see Tables VI–XII).

The algorithms are discussed in Section III-B. Numeric data types can originate from available data repositories or through remote sensing with IoT devices. Numeric values are easily processed and comprehended by various models. In comparison to image data types, numeric data types generally yield higher accuracy. However, one challenge with numerical data types is the presence of outliers, noise, and missing values, although these issues are not particularly concerning.

The most notable trend is the proliferation of ML algorithms tailored for prediction tasks, with LightGBM surpassing all others in this regard. LightGBM demonstrates proficiency in handling large datasets with numerous features. Although ANNs also excel in prediction tasks, issues such as overfitting and underfitting can lead to decreased accuracy during prediction. With ANNs, feature extraction is crucial prior to model training to enhance the visibility of patterns in data points to the ANN model.

In Fig. 3, the prediction accuracy of SVM is higher than its classification accuracy, indicating that SVM is more effective in prediction tasks compared to classification. Furthermore, the Gradient Boosting algorithm outperforms other algorithms in various tasks. These algorithms are effective due to several factors, such as handling nonlinearity, ensemble learning, sequential learning, and overfitting management (see Tables VI–XII).

IV. IMPLEMENTATION METHODOLOGY OF DIFFERENT RESEARCH PAPERS

Based on how they address the issue, papers can be divided into three categories.

TABLE VI
K MEAN FUNCTIONALITY IN PRECISION FARMING

Problem Description	Data Type	Methodology	Result	Ref
Identifying the potential corn-producing regions	Numeric	K-Means Algorithm enables researchers to analyze and cluster corn crop data effectively, providing valuable insights into the suitability of different areas for corn planting	K-Means clustering effectively categorizes the districts into two groups with distinct characteristics, providing valuable insights for agricultural planning and resource distribution	[46]
Predicting crop yield on farmland	Numeric	By integrating the clustering capabilities of K-means with the classification efficiency of Modified KNN, this article offers a comprehensive approach to predicting major crop yields.	The Methodology achieved an overall accuracy of 96%	[47]
Detecting and classifying pest	Image	The K-Means segmentation technique is essential for identifying pests in agricultural fields, as it segments images to detect pests or any other objects present.	KNN achieved high accuracy of 92.3% with effective feature extraction techniques, particularly DCT + PCA	[48]

TABLE VII
SVM FUNCTIONALITY IN PRECISION FARMING

Problem Description	Data Type	Methodology	Result	Ref
Monitoring corn under different weed control methods and nitrogen application rates	Image	SVM model used for classification of hyperspectral images which is based on the spectral reflectance values of the corn canopy.	Regular classification accuracy was 69.2%, 85.8%, and 80.8% for the effects of nitrogen and weeds combined, weeds alone, and nitrogen alone, respectively.	[49]
Classification of soil moisture into four class;	Numeric	The model is based on how the temperature of the soil and the atmosphere changes over the course of a day. The multiclassification model based on SVM employs the binary classification tree method.	Classification accuracy using a binary classification tree reached 82.9%.	[50]
Classification of five fruit based on shape and size	Image	Creation of an automated system that uses Fuzzy Logic and SVMs to grade and sort agricultural produce, respectively.	Considering the classification accuracy of five fruits, presented in this article the overall accuracy is 56.55% with 0% accuracy for carrot.	[51]
Detecting and classifying leaf disease	Image	Grid search is employed to identify the best or optimal hyperparameters for the SVM model, ensuring that the model achieves the highest level of prediction accuracy.	The SVM achieved an accuracy of 80%, while the grid search SVM achieved an accuracy of 84% on a dataset of 100 data points.	[52]
Predicting which crop is suitable to grow on an area based on four factors	Numeric	Soil, temperature, underground water, and rainfall are factors used to train linear SVM model for prediction	Prediction accuracy of 95.48% on the dataset.	[53]

- 1) Compiling information for smart systems (such as an irrigation system) monitoring, visualization, and operation.
- 2) Applying ML algorithms to pre-existing datasets.
- 3) Data Collection and utilizing ML algorithms on it.

A. Proposed Methodology

The novelty of the proposed methodology includes integrating ML with data collection (sensing). Sensors and data acquisition systems are utilized to gather data tracking system or environmental parameters. Subsequently, ML algorithms leverage this collected data as input. The data are then analyzed, processed,

and insightful predictions are derived using ML techniques as seen in Fig. 4.

1) *Data Acquisition*: In precision agriculture, sensors collect data on soil health, and ML algorithms optimize ideas of producing the best crop to grow based on the data.

The setup in Fig. 5(b) illustrates the design of IoT sensor for data collection of different soil and weather parameters. The data are collected and stored locally using Arduino Uno as a microcontroller.

Using RF algorithm (Fig. 9), a model was developed to recommend the best crop to grow based on nutrient levels collected by

TABLE VIII
DECISION TREE FUNCTIONALITY IN PRECISION FARMING

Problem Description	Data Type	Methodology	Result	Ref
Leaf disease detection and classification.	Image	Decision tree classifier is used to categories between healthy and unhealthy leaf.	Data set contains 1000 leaves, with classification accuracy of 96%.	[54]
Deciding when is suitable to water a crop.	Numeric	Sensors are used to collect temperature, humidity and moisture which are then fed to a trained decision tree model with binary output.	NA	[55]
Improve the accuracy of crop classification	Image	The decision tree algorithm utilized in this study identifies and refine inaccurate crop classifications in the Cropland Data Layer by incorporating spatial and temporal information.	Accuracy of 90%	[56]
Predict the yield of crops based on past yield value	Numeric	Ensembling of Decision Tree Regressor with Ada Boost Regressor to boost the weak learner in decision tree	Accuracy of 95.7% with data size of 28 242 instances	[57]

TABLE IX
RANDOMFOREST FUNCTIONALITY IN PRECISION FARMING

Problem Description	Data Type	Vegetation Indices	Methodology	Result	Ref
Predicting crop yield	Numeric	NA	The RF algorithm evaluates the features to determine if the crop is suitable for the agricultural field and provides the predicted crop yield as the output.	Model produced an accuracy greater than 95% compared to the Naive bayes algorithm with an accuracy of 80%	[58]
Maize yield prediction using multispectral UAV-based imagery	Image	NDVI	RF is utilized in this experiment to leverage multispectral UAV-based imagery and effectively predict maize yield by incorporating a ranking-based strategy for feature selection based on vegetation indices.	The model demonstrates the capability to account for more than 72% of the variance in maize yield, surpassing the performance of SVM and KNN.	[59]
Mapping irrigated areas	Image	NDVI	RF was utilized to automatically classify irrigated areas through satellite imagery, aiming to enhance the precision of mapping these regions within the study area.	The model achieves an 88% accuracy rate in classifying irrigated areas.	[60]
Carrot yield mapping in precision agriculture applications	Image	SAVI	RF algorithm is employed on a dataset that includes spectral data and verified yield samples to develop a model for predicting carrot yield.	Data set of 15093 observations model achieves R^2 value of 0.82	[61]

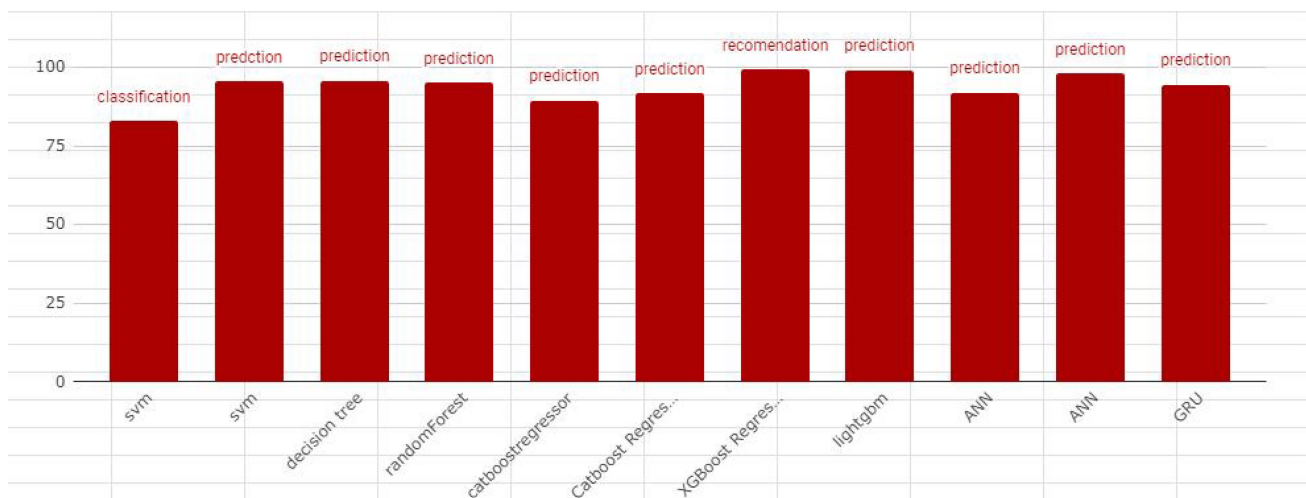


Fig. 3. Graph showing accuracy and purpose of ML algorithms that used data type "Numeric."

TABLE X
GRADIENT BOOSTING FUNCTIONALITY IN PRECISION FARMING

Problem Description	Algorithm	Data Type	Methodology	Result	Ref
Predicting the appropriate crop depending on weather forecasts.	Catboost Regressor	Numeric	Ten regression models are built using the weather forecast dataset.	The Elastic Net Regression exhibited the least performance, achieving R^2 value of 52.6%, while the CatBoost Regressor demonstrated the highest performance, reaching around 89.59%.	[62]
Predicting crop yield of three different crops	Catboost Regressor and XGBoost Regressor	Numeric	Developing an Ensemble ML Approach (EMLA) Model by combining the predictions of both algorithms from weather data.	Compared to 8 other boosting algorithms, EMLA excelled them all with R^2 values of 88.084%, 91.776%, and 90%.	[63]
Recommending crops in a given season and for a region depending on soil parameters.	XGBoost Regressor	Numeric	The XGBoost ML algorithm is applied to suggest the top three crops considering soil and weather conditions.	XGBoost has the highest accuracy of 99.3%, higher than RF, SVM, Decision Tree, Logistic Regression, Naive Bayes	[64]
Predicting Temperature in Greenhouse	LightGBM	Numeric	Extracting the time-series features of the factors affecting the indoor temperature in a greenhouse and then using the LightGBM model to more accurately predict the greenhouse temperature	LightGBM outperforms all the other models presented in the study with an R^2 value of 0.9905.	[65]
Recommending the best crop to grow based on soil nutrients	LightGBM	Numeric	CatBoostRegressor, XGBoostRegressor, and LightGBM models are employed to evaluate and categorize different predictive parameters.	-Catboost accuracy of 0.983 -XGBoost accuracy of 0.989 -LightGBM accuracy of 0.994	[66]

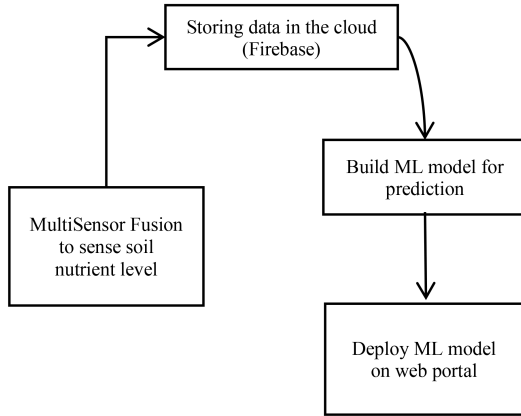


Fig. 4. Data collection from sensors, building ML model and performing correlation analysis with data stored in the cloud for flexibility, finally deploying ML model on the Internet.

the sensor. This approach transforms farming into a data-driven process informed by historical data rather than relying solely on traditional trial and error methods.

2) *Creating ML Model (RF)*: From historical data, an ML model is created in python using pandas and the sklearn library, with this model and the data collected as shown in Fig. 5(b), the model will be able to predict what sort of crop to produce based on soil nutrient level and environmental factors.

RF is an ensemble learning method for classification and regression tasks. It builds multiple decision trees during training and merges their prediction to improve the accuracy

and robustness

$$X_0 \left\{ \begin{array}{ll} X_1^{(rxc_i)} & \text{for } X_0 \neq 0 \\ X_2^{(rxc_{ii})} & \text{for } X_0 \neq 0 \\ X_{((n))}^{(rxc_{i+1})} & \text{for } X_0 \neq 0 \end{array} \right\}.$$

The original dataset (X_0) is split into subsets depending on whether the bootstrap parameter is set to True or False. With bootstrap = True, each tree in the forest is constructed using a bootstrapped sample of the dataset (X_0).

This process involves randomly drawing samples from the original dataset with replacement, effectively creating multiple subsets of the data. The size of these bootstrapped samples is controlled by the max_samples parameter

$$\forall X_1, X_2, \dots, X_{((n))} \in X_0.$$

Decision tree is constructed. A node will be split if this split induces a decrease of the impurity greater than or equal to this value

$$\Delta I = \frac{X_t}{X} \times \left(\text{impurity} - \frac{X_{tR}}{X_t} \times \text{right_impurity} - \frac{X_{tL}}{X_t} \times \text{left_impurity} \right) \quad (1)$$

where

X total number of samples;

X_t number of samples at the current node;

X_{tL} number of samples in the left child;

TABLE XI
DEEP LEARNING FUNCTIONALITY IN PRECISION FARMING

Problem Description	Data Type	Methodology	Result	Ref
Predicting what fertilizer is better for a given soil	Numeric	Developing an ANN model using feedforward and backpropagation techniques to recommend fertilizers according to the nutrient levels present in the soil.	50 test cases in total were taken into consideration; - 46 cases were correctly predicted. - 4 cases were incorrectly predicted; - 92% of the percentage was accurate.	[67]
Identifying Leaf spot in crops	Image	Images are segmented into clusters using K-means clustering, features such as contrast and homogeneity are extracted, and finally, an ANN classifier is used to classify the diseases.	The accuracy of disease detection for bacterial leaf spot and target spot of cotton leaf diseases is 90% and 80%, respectively. The accuracy was found to be 100% for tomato leaf diseases such as septoria leaf spot and leaf mold. It was discovered that the average classification accuracy for all four diseases was 92.5%.	[68]
Early detection and classification of plant diseases in crops using computer vision.	Image	The process involves image processing, feature extraction, classifier training, and performance evaluation to achieve accurate and efficient classification of plant diseases using SVM and ANN classifiers.	- Combining color and texture features with SVM resulted in an average accuracy of 98% while, ANN had an average accuracy of 87%	[69]
Creating an ANN based controller via MATLAB	Numeric	The parameters (air temperature, soil moisture, radiation, and humidity) are used to predict irrigation requirements, considering ecological factors, evapotranspiration, and crop varieties.	-Error between required and actual soil moisture remains steady and reasonable, typically less than 2%. -The ANN controller demonstrates low ON/OFF activity of the valve and energy system, resulting in significant savings in energy and water consumption.	[70]
Detecting leave diseases in corn	Image	The CNN model forecasts corn leaf disease using a processed dataset, incorporating image resizing, labeling, and splits for training. The model features three convolutional layers, ReLu activation, max-pooling, and the Adam optimizer, using dropout to prevent overfitting.	The model achieved an accuracy of 98.7% with shorter training times to reach optimal performance.	[71]
Crop yield prediction with satellite imagery	Image	With 3D convolutions, model is able to incorporate temporal information, taking into account both spatial features and the temporal signal. The LSTM component of the model is crucial for capturing temporal patterns and dependencies within the input data.	R^2 value of model was greater than 80% with MAPE value of 7.2% and 4.9% for maize and soybean field, respectively.	[72]
Classification of fruits	Image	Fruit images are captured with consistent lighting and processed to enhance quality. Features such as shape, color, and texture are extracted. These features feed into an ANN for classification.	This article achieved up to 96% accuracy, surpassing previous approaches, the technique integrates color and texture features.	[73]

X_{tR} number of samples in the right child;

$$\Delta I \geq \tau.$$

The decrease in impurity is due to the split at the current node. If this impurity decrease is greater than or equal to the threshold, then the node will be split.

3) *Voting*: The predicted class label for each particular sample $(X_1, X_2, \dots, X_{(n)})$ is the final predicted class label for a sample which is the class label that represents the majority (mode) of the class labels predicted by all the individual classifiers.

The RF aggregates the predictions of all classifiers and selects the class label that occurs most frequently among them as the final prediction for the sample

$$y'(X) = \text{mode}(f_1(X_1), f_2(X_2), \dots, f_T(X_{(\text{len}(\text{dataset})=0)})) \quad (2)$$

where

$f_t(X)$ prediction of tree t for input features X . Each tree t predicts a class label based on the input features;

TABLE XII
NEURAL NETWORK ARCHITECTURE FOR PRECISION FARMING

Problem Description	Algorithm	VI	Data Type	Methodology	Result	Ref
Predicting future weather forecast in IOT system	GRU	NA	Numeric	A two-level decomposition structure simplified weather data by breaking it into periods, while utilizing multiple GRUs to predict both long-term and medium-term forecasts effectively.	Model had MAE value of 97.24% and RMSE value of 94.43% as deviation from actual values are low	[74]
Predicting soil moisture at maize root level	CNN(Convulated Neural Network)-GRU	None	Image	Merging the feature extraction strengths of CNN with the time series memory capabilities of GRU to construct the hybrid CNN-GRU Model	The model's predictive accuracy increased as soil depth increased, with the hybrid CNN-GRU model achieving the highest R ² score of 0.97 at a depth of 60–80 cm.	[75]
Detecting soybean sudden death syndrome (SDS) disease development	GRU	NDVI	Image	GRU-based method is proposed to convert satellite images collected from PlanetScope into time series format, addressing the vanishing gradient problem encountered in standard RNNs when dealing with multivariate time-series imagery data.	Accuracy of GRU-based model was 82.5%–90.4%	[76]
leaf spot identification method with local-based fine-grained data augmentation	GAN	None	Image	The Fine Grained-GAN approach identifies Grape Leaf Spot under limited samples by augmenting the training dataset with synthetic spot images.	The recognition accuracy of five DL models was assessed, with particular attention to ResNet-50, which achieved an accuracy of 96.27%.	[77]

$\hat{y}(X)$ final prediction for input features X . It is the class label that occurs most frequently among the predictions of all trees, making it the predicted class label for the input X .

4) *Model and Data Analysis*: Now, calculating the uncertainty of RF prediction

$$\sigma_{\text{pred}} = \sqrt{\frac{1}{X} \sum_{i=1}^X (\hat{y}_i - \bar{y})^2}. \quad (3)$$

In Fig. 6, higher values on the y-axis signify increased uncertainty in predictions for the corresponding data points, while lower values indicate reduced uncertainty, indicating greater consistency among predictions from different trees.

This visualization helps in understanding the variability in predictions across various trees in the RF model, offering insights into the uncertainty linked with the predictions.

Also, analyzing the line chart of std of each data point in Fig. 7, we observe which features have higher variability compared to others. In Fig. 8, features with higher standard deviations have a more significant impact on the prediction model, while features with lower standard deviations have less influence.

The RF model is deployed on the Internet through the Flask framework, streamlining access to predictive analytics for farmers and enhancing agricultural decision-making. Seamlessly integrated into a user-friendly web application (Fig. 10), farmers input relevant data, such as weather conditions and crop types. After processing the data, the model generates precise predictions of the type of crop to produce. Results are presented on the interface, empowering farmers with actionable insights.

V. GENERAL ANALYSIS OF ENTIRE RESEARCH WORK DONE

A. Real-Life Application of Machine Learning Algorithm

Algorithms for ML are important for many kinds of research. The paper by Aravatagimath et al. [78] used ML in one instance to forecast and suggest appropriate crops for particular locations based on environmental parameters such as temperature, humidity, and rainfall. For crop prediction, the authors used a variety of algorithms, such as logistic regression, KNN, SVMs, RF, and naive Bayes classifier. They employed cooperative filtering and a multicondition filtering algorithm, an enhanced KNN algorithm renowned for its efficiency and ease of use without requiring preprocessing, to generate recommendations.

In order to estimate crop yields and related expenses, the article also examines database variations and the use of ML algorithms. In addition, in a different study [79], the K-means algorithm is used to classify agricultural products according to their prices in various regions. The major goal is to promote direct communication between farmers and customers, doing away with middleman's fees and establishing a win-win situation. One notable field of study is intelligent irrigation systems. A number of papers [80], [81], [82], [83] investigate smart farming systems with features such as temperature monitoring, cloud-based data management, and the ability to detect soil moisture and water levels. These systems provide dependable and affordable ways to maximize crop yields.

In a similar way, IoT is also used in garden irrigation, showing how it improves water usage by means of real-time soil moisture monitoring and remote pump control, which eventually results in effective garden management and better agricultural outcomes.

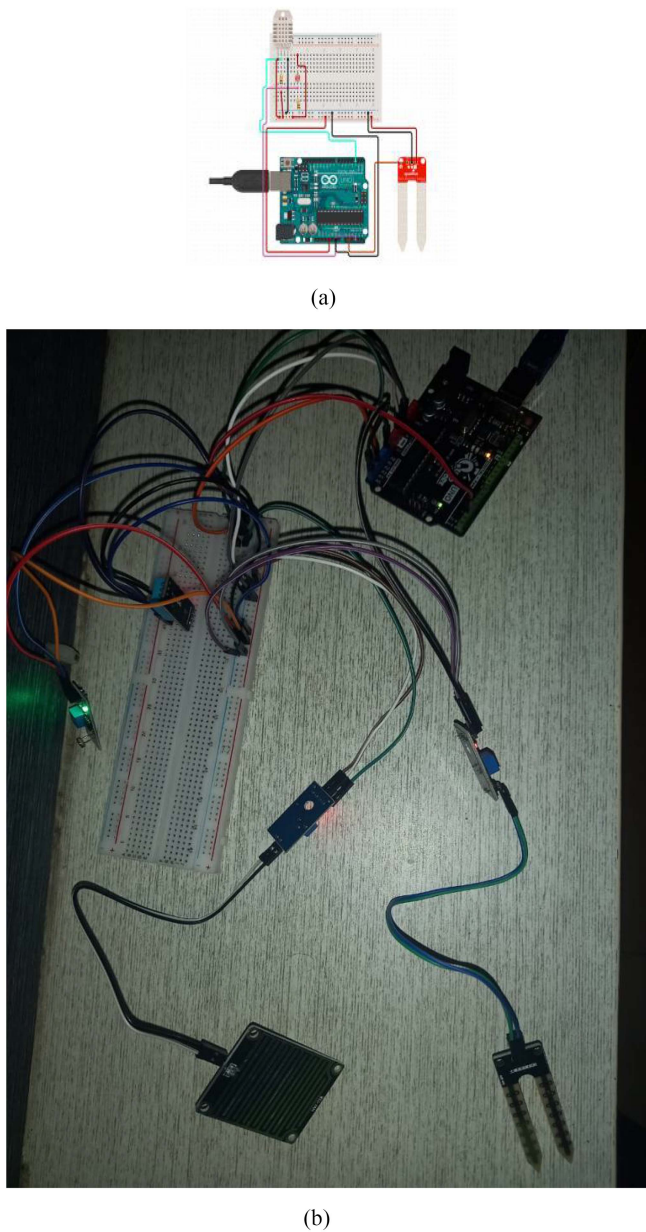


Fig. 5. (a) Circuit diagram of data collection with sensors. (b) Setup of IoT sensors ready for data collection.

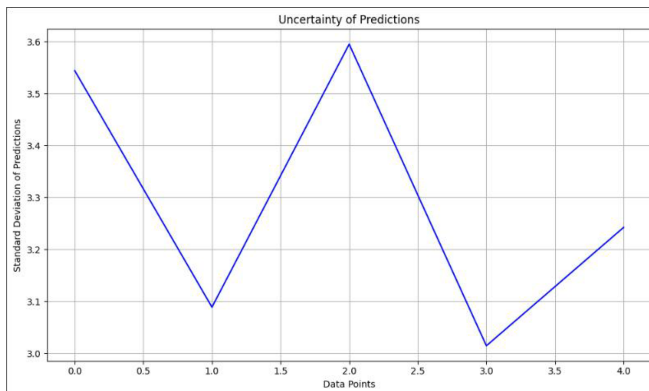


Fig. 6. Visualizing uncertainty of RF prediction after aggregating.

	Temperature	humidity	pH	N	P	K	Soil Moisture	Light Intensity
0	26.7000007	69	6.940086	82	145	31	35	369
1	27.7000007	73	7.738143	83	147	34	49	778
2	22.7000007	81	6.947932	87	152	26	46	385
3	24.7000007	67	8.271626	83	152	27	52	470
4	29.7999992	66	8.995152	84	148	21	41	369
5	29.7999992	77	6.809694	86	151	30	25	726
6	27.7999992	77	7.729002	85	150	34	26	515
7	28.8999996	75	9.346908	85	156	34	51	241
8	24.8999996	71	8.630803	88	156	30	46	796
9	23.8999996	73	7.136476	84	147	24	45	271
0	31.8999996	76	7.262192	83	149	29	36	1023
1	32.8999996	82	7.783147	80	150	34	31	339
2	22.8999996	78	8.553022	88	144	27	23	339
3	34.8999996	80	7.539952	88	154	24	68	762
4	33.7999992	77	7.062721	87	153	29	67	516
5	26.7999992	65	6.911119	86	148	29	66	1006
6	23.7999992	77	8.545509	86	150	32	66	339
7	29.7999992	76	6.923544	84	155	34	66	762
8	23.8999996	74	8.180189	82	144	31	68	516
9	22.8999996	75	7.107071	83	156	29	67	1006
0	22	74	8.251674	82	157	23	62	339
1	24	75	7.62331	79	151	25	70	762
2	32	72	7.531457	80	148	31	71	516
3	34.1000003	67	7.363141	88	154	19	69	1006
4	28.1000003	66	8.909858	89	152	22	74	515

Fig. 7. Spreadsheet of data collected from the sensors.

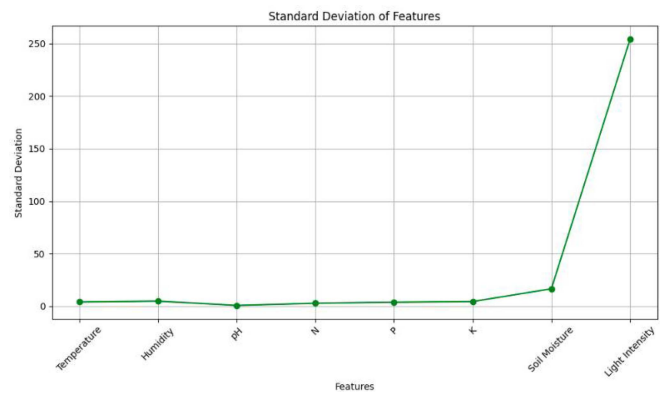


Fig. 8. Standard deviation of dataset.

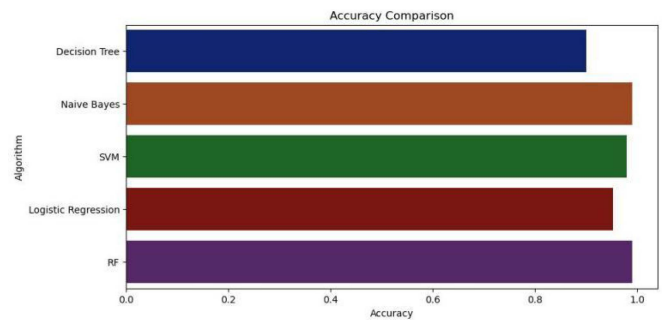


Fig. 9. Evaluation metrics of the most suitable algorithm with RF having the highest accuracy (0.98) of all other algorithms.

B. Emerging Technologies

1) *Satellite Imagery*: Utilizing satellite imagery is a useful method for assessing pasture land. It offers data and photos in high resolution, resulting in insightful conclusions. Multispectral band images from Landsat-8 were obtained and processed for the aforementioned study [84]. By using remote sensing indices [vegetation index (VI)], crop growth and yield are connected to these images. K-means and Mask R-CNN techniques were used in both Python and Matlab to evaluate changes in crop growth over time. In order to locate paddy fields at the pixel level, researchers used an advanced spatiotemporal-spectral deep neural network in conjunction with a multitemporal high-spatial

Nitrogen
80

Phosphorous
45

Potassium
60

Temperature
28

Humidity
70.3

pH level
7.0

Soil Moisture
27

Light Intensity
567

Predict

You should grow *banana* in your farm

Fig. 10. Web portal to predict the type of crop to grow.

resolution classification method to create an autonomous system that could distinguish between crop and noncrop areas [85].

The paper [86] presented a method for accurate agricultural area monitoring with a particular focus on identifying sparse and dense areas in sugarcane fields. Utilizing a blend of drone imagery and Landsat 8 satellite data, the research suggests an adaptive classification technique to lessen drone usage that occurs repeatedly. The process includes gathering ground truth data, segmenting drone images, and masking clouds. For classification, the short-wave infrared 1 spectral band is chosen, and a multiobjective optimization strategy is used to apply an adaptive thresholding technique. The results show that this affordable solution has the potential to be used for large-scale precision agriculture monitoring. Approximately 87% of the testing data and 73% of the validation data demonstrate high accuracy.

2) *Drone Technology*: The agricultural sector has undergone a revolution, thanks to drone technology, which has given farmers new tools to maximize their productivity. Precision farming, field mapping, crop spraying, and data collection are among the applications of drones [23]. They can assist farmers in raising yield rates, enhancing crop quality, and lowering operating expenses. Drone farming is a big step toward environmentally friendly farming methods and can boost productivity while preserving the environment. With a mounted camera, the drone automatically takes pictures of the leaves. Since hexacopters are lighter and more durable than octocopters but less stable than quadcopters, the system makes use of them [87].

Drone technology is also used to spray crops to prevent harmful chemicals from coming into contact with people [88]. Small drones are outfitted with tanks to carry fertilizer and pesticide sprayers as a part of the UAV (drone) for fertilizer and pesticide spraying design. The drones are in charge of monitoring the environment and agricultural fields, as well as applying chemical sprays to crops. The low-volume spraying techniques that the drones are intended to use prevent water waste and contamination of ground waters. It is possible to

extend both the drone's battery life and its maximum carrying capacity of fertilizer.

Equipped with advanced sensors and high-resolution cameras, drones provide farmers with real-time imagery for assessing crop health, identifying potential threats, and optimizing management strategies [89], [90]. This capability serves as a cornerstone for informed decision-making, ultimately leading to increased crop yields and enhanced farm productivity. Conducting aerial surveys, drones offer farmers crucial insights into the topography, soil health, and drainage patterns, facilitating strategic decision-making in terms of planting locations, seeding rates, and irrigation planning [91].

Advanced sensors analyze plant color and health, enabling drones to assess nitrogen levels with precision. This data-driven approach allows farmers to apply nitrogen fertilizers more judiciously, minimizing environmental impact and contributing to improved crop quality and yield [92]. Also, equipped with multi-spectral imaging capabilities, drones assess crop health throughout the growing season, enabling farmers to make timely adjustments to management practices, optimize harvest planning, and identify potential issues affecting overall productivity [93].

In the cases of drought assessment, drone aerial surveys capture crucial data on soil moisture levels and crop stress indicators, empowering farmers to implement water management strategies and optimize irrigation practices to mitigate the impact of drought on crop yields [94].

In orchards and agroforestry settings, drones shine in their ability to classify trees. Utilizing advanced imaging technology, drones distinguish between tree species, assess their health, and identify areas requiring attention. This capability enhances orchard management by enabling targeted interventions, such as pest control or disease management, based on specific tree classifications [95].

VI. COMMON CHALLENGES IN INTEGRATING IOT SENSORS, CLOUD PLATFORMS, AND MACHINE LEARNING ALGORITHMS

- 1) The network uses too many nodes, resulting in high costs and low power consumption.
- 2) There is a need for more diverse datasets, but collecting and labeling a large amount of data is challenging.
- 3) It is important to control costs, and the current database system may not handle real-time processing of the massive data influx in IoT, which involves a large number of nodes.
- 4) Issues in the sensor cloud architecture include authorization, energy efficiency, security, storage, and network management.
- 5) The most challenging phase in deploying models is due to the lack of deployment skills, third-party library dependencies, complex real-world scenarios, and limitations of the deployment platform's hardware.
- 6) Finding the most suitable algorithm to create a customized ML model is also a challenge.

A. Policy and Regulatory Landscape for IoT and Cloud Adoption in Agriculture

- 1) Ensuring data privacy and security is crucial to building trust and confidence in IoT systems among farmers and

stakeholders. Implementing robust security regulations and standards can mitigate risks associated with hacking and unethical data collection practices.

- 2) Establishing standards for data exchange and interoperability is essential for the seamless integration of IoT systems across the agricultural value chain. This promotes efficiency and collaboration among different entities and systems involved in agri-food production.
- 3) Investing in improving network infrastructure, particularly in rural areas, is vital to overcome challenges related to connectivity and ensure reliable communication for IoT devices deployed in agricultural settings.
- 4) Setting energy efficiency standards for IoT devices helps address concerns about power consumption and battery life, especially in remote agricultural locations where access to electricity may be limited.
- 5) Developing clear regulations and frameworks for data governance helps address concerns about data ownership, usage rights, and protection, fostering trust and transparency in IoT applications in agriculture.
- 6) Promoting awareness and providing education about the benefits and applications of IoT technologies in agriculture is essential for encouraging adoption and acceptance among farmers and end-users.
- 7) Establishing usability standards for IoT solutions ensures that they are user-friendly and accessible to stakeholders with varying levels of technical expertise, facilitating adoption and reducing barriers to entry [96].

B. Research Gap

The first research gap, as identified in the paper by Sharma et al. [97], underscores the necessity for more extensive studies on the utilization of ML and IoT in agriculture. The gap in understanding lies in how these technologies can be effectively integrated and scaled in diverse agricultural contexts.

In the second research gap highlighted by Sasi Bhanu et al. [98], there is an emphasis on the need for further exploration of smart agriculture to unlock its full potential. The current lack of research on how predictive analysis can be systematically employed to enhance agricultural practices suggests a gap in our understanding of the full spectrum of possibilities that predictive analytics holds for optimizing and revolutionizing agricultural processes.

Finally, it is important to combine smart agriculture with operation research theories. There is a need to explore this more because we do not fully understand the benefits and improvements that can come from merging these two areas. This shows that there is a lot of potential for future research in this field.

VII. CONCLUSION

This article provides a thorough analysis of current agricultural research initiatives, emphasizing the utilization of ML algorithms, IoT sensors, and integration with cloud technology to address crop production uncertainties. Modern agricultural practices are shaped by technological advancements, as evidenced by the investigation of network technologies such as LoRa,

Wi-Fi, and Zigbee for data transmission, and the critical role of cloud technology in processing and analyzing IoT sensor data. Furthermore, This article identifies the technological, social, economic, and organizational factors that affect IoT adoption. The potential of ML algorithms for predictive modeling and disease detection is demonstrated through thorough research into these algorithms. In addition, the article provides a use case of IoT and ML to predict the best crop to cultivate based on soil nutrients and incorporates emerging technologies, such as drones and satellite imagery, expanding the application of precision agriculture. Moreover, the article provides insightful information into future trends and explores the policies and regulatory landscape for IoT and cloud adoption in agriculture. All things considered, this article is a useful tool that offers an integrated perspective of the state of the art, as well as areas that are ready for more investigation in the ever-evolving field of agricultural research.

ACKNOWLEDGMENT

Authors' contribution: The guide of this research was A. K. Singh; the creation of RandomForest model for the proposed methodology was by B. Abdoul-Rahmane; IOT setup for the proposed methodology was done by N. L. Mainsah; and finally, F. N. Fru was in charge of presenting the different ideas in the research.

REFERENCES

- [1] A. Tagarakis, V. Liakos, L. Perlepes, S. Fountas, and T. Gemtos, "Wireless sensor network for precision agriculture," in *Proc. 15th Panhellenic Conf. Inform.*, Kastoria, Greece, 2011, pp. 397–402, doi: [10.1109/PCI.2011.60](https://doi.org/10.1109/PCI.2011.60).
- [2] S. Puengsungwan, "IoT based soil moisture sensor for smart farming," in *Proc. Int. Conf. Power, Energy Innov.*, Oct. 2020, pp. 221–224.
- [3] Z. Hong, Z. Kalbarczyk, and R. K. Iyer, "A data-driven approach to soil moisture collection and prediction," in *Proc. IEEE Int. Conf. Smart Comput.*, 2016, pp. 1–6, doi: [10.1109/smartcomp.2016.750167](https://doi.org/10.1109/smartcomp.2016.750167).
- [4] A. Nayyar and V. Puri, "Smart farming: IoT based smart sensors agriculture stick for live temperature and moisture monitoring using Arduino, cloud computing & solar technology," in *Proc. IEEE Int. Conf. Commun. Comput. Syst.*, Sep. 2016, pp. 9781315364094–9781315364121.
- [5] J. Doshi, T. Patel, and S. kumar Bharti, "Smart Farming using IoT, a solution for optimally monitoring farming conditions," *Procedia Comput. Sci.*, vol. 160, pp. 746–751, 2019.
- [6] P. Maiti, J. Shukla, B. Sahoo, and A. Turuk, "Efficient data collection for IoT services in edge computing environment," in *Proc. Int. Conf. Inf. Technol.*, 2017, pp. 101–106, doi: [10.1109/ICIT.2017.40](https://doi.org/10.1109/ICIT.2017.40).
- [7] X. Li, L. Zhu, X. Chu, and H. Fu, "Edge computing-enabled wireless sensor networks for multiple data collection tasks in smart agriculture," *J. Sensors*, vol. 2020, no. 1, 2020, Art. no. 4398061.
- [8] A. Kumar, "Smart agriculture using IoT," *Int. J. Innov. Res. Comput. Sci. Technol.*, vol. 9, no. 6, pp. 188–191, 2021.
- [9] K. Raviteja and M. Supriya, "IoT-based agriculture monitoring system," in *Proc. 3rd Data Eng. Commun. Technol.*, 2020, pp. 473–483.
- [10] D. N. Venu, "Smart agriculture remote monitoring system using low power IOT network," *Int. J. Food Nutr. Sci.*, vol. 11, no. 6, pp. 327–340, 2022.
- [11] S. Gutiérrez, I. Martínez, J. Varona, M. Cardona, and R. Espinosa, "Smart mobile LoRa agriculture system based on Internet of Things," in *Proc. IEEE 39th Central Amer. Panama Conv.*, Guatemala City, Guatemala, 2019, pp. 1–6, doi: [10.1109/CONCAPANXXXIX47272.2019.8977109](https://doi.org/10.1109/CONCAPANXXXIX47272.2019.8977109).
- [12] S. J. S. Prasad et al., "An efficient LoRa-based smart agriculture management and monitoring system using wireless sensor networks," *Int. J. Ambient Energy*, vol. 43, no. 1, pp. 5447–5450, 2022, doi: [10.1080/01430750.2021.1953591](https://doi.org/10.1080/01430750.2021.1953591).
- [13] E. Kökten, B. C. Çalışkan, S. Karamzadeh, and E. G. Soyak, "Low-powered agriculture IoT systems with LoRa," in *Proc. IEEE Microw. Theory Techn. Wireless Commun.*, Riga, Latvia, 2020, pp. 178–183, doi: [10.1109/MTTW51045.2020.9244927](https://doi.org/10.1109/MTTW51045.2020.9244927).

- [14] W. Chanwattanapong, S. Hongdumnuen, B. Kumkhet, S. Junon, and P. Sangmahamad, "LoRa network based multi-wireless sensor nodes and LoRa gateway for agriculture application," in *Proc. Res., Invent., Innov. Congr., Innov. Elect. Electron.*, Sep. 2021, pp. 133–136.
- [15] M. Jiménez-Buendía, F. Soto-Valles, P. J. Blaya-Ros, A. Toledo-Moreo, R. Domingo-Miguel, and R. Torres-Sánchez, "High-density Wi-Fi based sensor network for efficient irrigation management in precision agriculture," *Appl. Sci.*, vol. 11, no. 4, 2021, Art. no. 1628.
- [16] L. García, S. Viciano-Tudela, S. Sendra, and J. Lloret, "Practical design of a Wi-Fi-based wireless sensor network for precision agriculture in citrus crops," in *Proc. 19th Int. Conf. Wireless Netw. Mobile Syst.*, Jul. 2022, pp. 107–114.
- [17] V. Novák, M. Stočes, E. Kánská, J. Pavlík, and J. Jarolímek, "Monitoring of movement on the farm using WiFi technology," *AGRI-Line Papers Econ. Inform.*, vol. 11, no. 4, pp. 85–92, 2019.
- [18] B. N. Alhasnawi, B. H. Jasim, and B. A. Issa, "Internet of Things (IoT) for smart precision agriculture," *Iraqi J. Elect. Electron. Eng.*, vol. 16, no. 1, pp. 28–38, 2020.
- [19] M. A. T. Tran, T. N. Le, and T. P. Vo, "Smart-Config WiFi technology using ESP8266 for low-cost wireless sensor networks," in *Proc. Int. Conf. Adv. Comput. Appl.*, Ho Chi Minh City, Vietnam, 2018, pp. 22–28, doi: [10.1109/ACOMP.2018.00012](https://doi.org/10.1109/ACOMP.2018.00012).
- [20] A. U. H. Hashmi et al., "Effects of IoT communication protocols for precision agriculture in outdoor environments," *IEEE Access*, vol. 12, pp. 46410–46421, 2024.
- [21] R. Tang, N. K. Aridas, and M. S. A. Talip, "Design of wireless sensor network for agricultural greenhouse based on improved Zigbee protocol," *Agriculture*, vol. 13, no. 8, 2023, Art. no. 1518.
- [22] O. Ayurzana and S. Tsagaanchuluun, "Monitoring system of agriculture fields using ZigBee modules," *Int. J. Adv. Smart Convergence*, vol. 10, no. 1, pp. 89–96, 2021.
- [23] F. Bu and X. Wang, "A smart agriculture IoT system based on deep reinforcement learning," *Future Gener. Comput. Syst.*, vol. 99, pp. 500–507, Oct. 1, 2019, doi: [10.1016/j.future.2019.04.041](https://doi.org/10.1016/j.future.2019.04.041).
- [24] M. S. Mekala and P. Viswanathan, "A Survey: Smart agriculture IoT with cloud computing," in *Proc. Int. Conf. Microelectron. Devices, Circuits Syst.*, Vellore, India, 2017, pp. 1–7, doi: [10.1109/ICMDCS.2017.8211551](https://doi.org/10.1109/ICMDCS.2017.8211551).
- [25] M. Dholu and K. A. Ghodinde, "Internet of Things (IoT) for precision agriculture application," in *Proc. 2nd Int. Conf. Trends Electron. Inform.*, Tirunelveli, India, 2018, pp. 339–342, doi: [10.1109/ICOEI.2018.8553720](https://doi.org/10.1109/ICOEI.2018.8553720).
- [26] S. Rajeswari, K. Suthendran, and K. Rajakumar, "A smart agricultural model by integrating IoT, mobile and cloud-based Big Data analytics," in *Proc. Int. Conf. Intell. Comput. Control*, Coimbatore, India, 2017, pp. 1–5, doi: [10.1109/I2C2.2017.8321902](https://doi.org/10.1109/I2C2.2017.8321902).
- [27] J. Das V. S. Sharma, and A. Kaushik, "Views of Irish farmers on smart farming technologies: An observational study," *AgriEngineering*, vol. 1, no. 2, pp. 164–187, 2019.
- [28] J. Singh and M. Kumari, "Socio-economic impacts of agricultural technologies on farmers' livelihood," *Nat. Acad. Sci. Lett.*, vol. 47, no. 3, pp. 227–231, 2024.
- [29] D. S. Gangwar, S. Tyagi, and S. K. Soni, "A techno-economic analysis of digital agriculture services: An ecological approach toward green growth," *Int. J. Environ. Sci. Technol.*, vol. 19, no. 5, pp. 3859–3870, 2022.
- [30] S. Misra, S. Chatterjee, and M. S. Obaidat, "On theoretical modeling of sensor cloud: A paradigm shift from wireless sensor network," *IEEE Syst. J.*, vol. 11, no. 2, pp. 1084–1093, Jun. 2017, doi: [10.1109/JSYST.2014.2362617](https://doi.org/10.1109/JSYST.2014.2362617).
- [31] R. Katarya, A. Raturi, A. Mehndiratta, and A. Thapper, "Impact of machine learning techniques in precision agriculture," in *Proc. 3rd Int. Conf. Emerg. Technol. Comput. Eng., Mach. Learn. Internet Things*, Jaipur, India, 2020, pp. 1–6, doi: [10.1109/ICETCE48199.2020.9091741](https://doi.org/10.1109/ICETCE48199.2020.9091741).
- [32] R. N. Bhimanpallewar and M. R. Narasingarao, "Alternative approaches of machine learning for agriculture advisory system," in *Proc. 10th Int. Conf. Cloud Comput., Data Sci. Eng.*, Noida, India, 2020, pp. 27–31, doi: [10.1109/Confluence47617.2020.9058152](https://doi.org/10.1109/Confluence47617.2020.9058152).
- [33] V. Sellam and E. Poovammal, "Prediction of crop yield using regression analysis," *Indian J. Sci. Technol.*, vol. 9, pp. 1–5, 2016.
- [34] A. Agrawal, D. Arora, and P. Sharma, "Crop production prediction using machine learning: An Indian perspective," in *Proc. Emerg. Technol. Data Mining Inf. Secur.*, 2021, vol. 3, pp. 981–990.
- [35] H. M. Ruiz-Juárez, J. Castillo-Araujo, M. G. Orozco-del-Castillo, N. L. Cuevas-Cuevas, F. Cárdenas-Pimentel, and R. Cáceres-Escalante, "A decision tree as an explainable artificial intelligence technique for identifying agricultural production predictor variables in Mexico," in *Proc. Int. Congr. Telematics Comput.*, Nov. 2023, pp. 1–14.
- [36] M. B. G. Bhavani, M. G. Kumar, M. Lakshmi, M. KNVPS, and B. Ramesh, "Prediction of various crops in agricultural field using decision tree and naive bayes algorithm in machine learning," *Int. J. Eng. Res. Technol.*, vol. 9, pp. 79–83, 2021.
- [37] J. Lu, Y. Liu, and X. Li, "The decision tree application in agricultural development," in *Proc. Artif. Intell. Comput. Intell., Lecture Notes Comput. Sci.*, H. Deng, D. Miao, J. Lei, and F. L. Wang, Eds., Berlin, Germany, Springer, 2011, vol. 7002, pp. 372–379, doi: [10.1007/978-3-642-23881-9_49](https://doi.org/10.1007/978-3-642-23881-9_49).
- [38] N. Prasath, J. Sreemathy, N. Krishnaraj, and P. Vigneshwaran, "Analysis of crop yield prediction using random forest regression model," in *Proc. Inf. Syst. Intell. Syst.*, 2023, pp. 239–249.
- [39] S. Data, "Crop Yield estimation using decision trees and random forest machine learning algorithms on data from terra," *Mach. Learn. Data Mining Aerosp. Technol.*, vol. 836, 2019, Art. no. 107.
- [40] J. H. Jeong et al., "Random forests for global and regional crop yield predictions," *PLoS One*, vol. 11, no. 6, 2016, Art. no. e0156571.
- [41] T. K. Mishra, S. K. Mishra, K. J. Sai, B. S. Alekhyia, and A. R. Nishith, "Crop recommendation system using KNN and random forest considering Indian data set," in *Proc. 19th OITS Int. Conf. Inf. Technol.*, Dec. 2021, pp. 308–312.
- [42] G. K. Kumar et al., "Internet of things sensors and support vector machine integrated intelligent irrigation system for agriculture industry," *Discover Sustainability*, vol. 5, no. 1, 2024, Art. no. 6.
- [43] T. K. Mishra, S. K. Mishra, K. J. Sai, S. Peddi, and M. Surusomayajula, "Crop recommendation system using support vector machine considering Indian dataset," in *Proc. Adv. Distrib. Comput. Mach. Learn.*, 2022, pp. 501–510.
- [44] R. K. Paul et al., "Machine learning techniques for forecasting agricultural prices: A case of brinjal in Odisha India," *PLoS One*, vol. 17, no. 7, 2022, Art. no. e0270553.
- [45] M. R. Sundari, G. S. R. Krishna, V. S. Naveen, and G. Bharathi, "Crop recommendation system using k-nearest neighbors algorithm," in *Proc. 6th Int. Conf. Recent Trends Comput.*, 2021, pp. 581–589.
- [46] A. Sharma, A. Jain, P. Gupta, and V. Chowdary, "Machine learning applications for precision agriculture: A comprehensive review," *IEEE Access*, vol. 9, pp. 4843–4873, 2021.
- [47] A. Suresh, P. G. Kumar, and M. Ramalatha, "Prediction of major crop yields of Tamil Nadu using K-means and modified KNN," in *Proc. 3rd Int. Conf. Commun. Electron. Syst.*, Oct. 2018, pp. 88–93.
- [48] M. H. Javed, M. Humair, B. Yaqoob, N. Noor, and T. Arshad, "K-means based automatic pests detection and classification for pesticides spraying," *Int. J. Adv. Comput. Sci. Appl.*, vol. 8, pp. 236–240, 2017, doi: [10.14569/IJACSA.2017.081131](https://doi.org/10.14569/IJACSA.2017.081131).
- [49] Y. Karimi, S. O. Prasher, R. M. Patel, and S. H. Kim, "Application of support vector machine technology for Weed and nitrogen stress detection in corn," *Comput. Electron. Agriculture*, vol. 51, no. 1/2, pp. 99–109, 2006.
- [50] X. Gao, T. Lu, P. Liu, and Q. Lu, "A soil moisture classification model based on SVM used in agricultural WSN," in *Proc. IEEE 7th Joint Int. Inf. Technol. Artif. Intell. Conf.*, Chongqing, China, 2014, pp. 432–436, doi: [10.1109/ITAIC.2014.7065086](https://doi.org/10.1109/ITAIC.2014.7065086).
- [51] N. B. A. Mustafa, S. K. Ahmed, Z. Ali, W. B. Yit, A. A. Z. Abidin, and Z. A. M. Sharrif, "Agricultural produce sorting and grading using support vector machines and fuzzy logic," in *Proc. IEEE Int. Conf. Signal Image Process. Appl.*, Kuala Lumpur, Malaysia, 2009, pp. 391–396, doi: [10.1109/ICSIPA.2009.5478684](https://doi.org/10.1109/ICSIPA.2009.5478684).
- [52] M. Bhagat, D. Kumar, I. Haque, H. S. Munda, and R. Bhagat, "Plant leaf disease classification using grid search based SVM," in *Proc. 2nd Int. Conf. Data, Eng. Appl.*, Bhopal, India, 2020, pp. 1–6, doi: [10.1109/IDEA49133.2020.9170725](https://doi.org/10.1109/IDEA49133.2020.9170725).
- [53] V. K. Gunjan and J. M. Zurada, Eds., "Prediction of agriculture yields using machine learning algorithms," in *Proc. 2nd Int. Conf. Recent Trends Mach. Learn., IoT, Smart Cities Appl.* (Lecture Notes in Networks and Systems Series), Singapore, Springer, 2022, vol. 237, pp. 17–26, doi: [10.1007/978-981-16-6407-6_2](https://doi.org/10.1007/978-981-16-6407-6_2).
- [54] B. Rajesh, M. V. Sai Vardhan, and L. Sujihelen, "Leaf disease detection and classification by decision tree," in *Proc. 4th Int. Conf. Trends Electron. Inform.*, Tirunelveli, India, 2020, pp. 705–708, doi: [10.1109/ICOEI48184.2020.9142988](https://doi.org/10.1109/ICOEI48184.2020.9142988).
- [55] K. S. P. Reddy, Y. M. Roopa, K. R. LN, and N. S. Nandan, "IoT based smart agriculture using machine learning," in *Proc. 2nd Int. Conf. Inventive Res. Comput. Appl.*, Jul. 2020, pp. 130–134.
- [56] L. Lin et al., "Validation and refinement of cropland data layer using a spatial-temporal decision tree algorithm," *Sci. Data*, vol. 9, no. 1, 2022, Art. no. 63.

- [57] M. Keerthana, K. J. M. Meghana, S. Pravallika, and M. Kavitha, "An ensemble algorithm for crop yield prediction," in *Proc. 3rd Int. Conf. Intell. Commun. Technol. Virtual Mobile Netw.*, Tirunelveli, India, 2021, pp. 963–970, doi: [10.1109/ICICV50876.2021.9388479](https://doi.org/10.1109/ICICV50876.2021.9388479).
- [58] V. Geetha, A. Punitha, M. Abarna, M. Akshaya, S. Illakiya, and A. P. Janani, "An effective crop prediction using random forest algorithm," in *Proc. Int. Conf. Syst., Comput., Automat. Netw.*, Pondicherry, India, 2020, pp. 1–5, doi: [10.1109/ICSCAN49426.2020.9262311](https://doi.org/10.1109/ICSCAN49426.2020.9262311).
- [59] A. Islam, I. Khair, S. Hossain, R. A. Ifty, M. N. Arefin, and M. J. Patwary, "Ensemble machine learning approach for agricultural crop selection," in *Proc. Int. Conf. Elect., Comput. Commun. Eng.*, Feb. 2023, pp. 1–6.
- [60] A. Islam, I. Khair, R. A. Ifty, S. Hossain, M. N. Arefin, and F. B. Al Abid, "Bangladeshi agricultural product forecasting approach using permutation driven machine learning algorithm," in *Proc. Int. Conf. Recent Progresses Sci., Eng. Technol.*, Dec. 2022, pp. 1–6.
- [61] A. P. M. Ramos et al., "A random forest ranking approach to predict yield in maize with UAV-based vegetation spectral indices," *Comput. Electron. Agriculture*, vol. 178, 2020, Art. no. 105791.
- [62] J. Magidi, L. Nhamo, S. Mpandeli, and T. Mabhaudhi, "Application of the random forest classifier to map irrigated areas using Google Earth Engine," *Remote Sens.*, vol. 13, no. 5, 2021, Art. no. 876.
- [63] M. Deshmukh, A. Jaiswar, O. Joshi, and R. Shedje, "Farming assistance for soil fertility improvement and crop prediction using XGBoost," presented at the Proc. ITM Web Conf., 2022, vol. 44, Paper 03022.
- [64] M. C. F. Wei, L. F. Maldaner, P. M. N. Ottoni, and J. P. Molin, "Carrot yield mapping: A precision agriculture approach based on machine learning," *AI*, vol. 1, no. 2, pp. 229–241, 2020.
- [65] Q. Cao, Y. Wu, J. Yang, and J. Yin, "Greenhouse temperature prediction based on time-series features and LightGBM," *Appl. Sci.*, vol. 13, no. 3, 2023, Art. no. 1610.
- [66] M. Ackson, G. Marvin, and A. Chakrabarty, "Robust ensemble machine learning for precision agriculture," in *Proc. Int. Conf. Innov. Sci., Eng. Technol.*, Feb. 2022, pp. 1–6.
- [67] G. Ravichandran and R. S. Koteeshwari, "Agricultural crop predictor and advisor using ANN for smartphones," in *Proc. Int. Conf. Emerg. Trends Eng., Technol. Sci.*, Pudukkottai, India, 2016, pp. 1–6, doi: [10.1109/ICETETS.2016.7603053](https://doi.org/10.1109/ICETETS.2016.7603053).
- [68] C. U. Kumari, S. Jeevan Prasad, and G. Mounika, "Leaf disease detection: Feature extraction with K-means clustering and classification with ANN," in *Proc. 3rd Int. Conf. Comput. Methodol. Commun.*, Erode, India, 2019, pp. 1095–1098, doi: [10.1109/ICCMC.2019.8819750](https://doi.org/10.1109/ICCMC.2019.8819750).
- [69] D. Pujari, R. Yakkundimath, and A. S. Byadgi, "SVM and ANN based classification of plant diseases using feature reduction technique," *IJIMAI*, vol. 3, no. 7, pp. 6–14, 2016.
- [70] S. M. Umair and R. Usman, "Automation of irrigation system using ANN based controller," *Int. J. Elect. Comput. Sci. IJECS-IJENS*, vol. 10, no. 2, pp. 41–47, 2010.
- [71] B. Pratap, N. Agarwal, and S. Joshi, "Development of ANN based efficient fruit recognition technique," *Glob. J. Comput. Sci. Technol.*, vol. 14, no. C5, pp. 1–6, 2014.
- [72] K. P. Panigrahi, A. K. Sahoo, and H. Das, "A CNN approach for corn leaves disease detection to support digital agricultural system," in *Proc. 4th Int. Conf. Trends Electron. Inform.*, Tirunelveli, India, 2020, pp. 678–683, doi: [10.1109/ICOEI48184.2020.9142871](https://doi.org/10.1109/ICOEI48184.2020.9142871).
- [73] X. B. Jin, X. H. Yu, X. Y. Wang, Y. T. Bai, T. L. Su, and J. L. Kong, "Deep learning predictor for sustainable precision agriculture based on Internet of Things system," *Sustainability*, vol. 12, no. 4, 2020, Art. no. 1433.
- [74] J. Yu, X. Zhang, L. Xu, J. Dong, and L. Zhangzhong, "A hybrid CNN-GRU model for predicting soil moisture in maize root zone," *Agricultural Water Manage.*, vol. 245, 2021, Art. no. 106649.
- [75] L. Bi, G. Hu, M. M. Raza, Y. Kandel, L. Leandro, and D. Mueller, "A gated recurrent units (GRU)-based model for early detection of soybean sudden death syndrome through time-series satellite imagery," *Remote Sens.*, vol. 12, no. 21, 2020, Art. no. 3621.
- [76] C. Zhou, Z. Zhang, S. Zhou, J. Xing, Q. Wu, and J. Song, "Grape leaf spot identification under limited samples by fine grained-GAN," *IEEE Access*, vol. 9, pp. 100480–100489, 2021, doi: [10.1109/ACCESS.2021.3097050](https://doi.org/10.1109/ACCESS.2021.3097050).
- [77] S. Talukder, H. Jannat, K. Sengupta, S. Saha, and M. I. Hossain, "Enhancing crops production based on environmental status using machine learning techniques," in *Proc. Int. Conf. Comput. Sci. Its Appl. Agriculture*, Bogor, Indonesia, 2020, pp. 1–5, doi: [10.1109/ICOSICA49951.2020.9243251](https://doi.org/10.1109/ICOSICA49951.2020.9243251).
- [78] A. Aravatagimath, A. V. Sutagundar, and V. Yalavigi, "Agriculture product marketing data analysis using machine learning," in *Proc. Int. Conf. Forensics, Analytics, Big Data, Secur.*, Bengaluru, India, 2021, pp. 1–6, doi: [10.1109/FABS52071.2021.9702674](https://doi.org/10.1109/FABS52071.2021.9702674).
- [79] M. B. Tephila, R. A. Sri, R. Abinaya, J. A. Lakshmi, and V. Divya, "Automated smart irrigation system using IoT with sensor parameter," in *Proc. Int. Conf. Electron. Renewable Syst.*, Tuticorin, India, 2022, pp. 543–549, doi: [10.1109/ICEARS53579.2022.9751993](https://doi.org/10.1109/ICEARS53579.2022.9751993).
- [80] R. Vijay, Y. V. Priya, P. C. Reddy, S. Monisha, and V. Ramasamy, "A IoT based smart auto irrigation system," in *Proc. Int. Conf. Comput. Commun. Inform.*, Coimbatore, India, 2023, pp. 1–4, doi: [10.1109/IC-CCIS56745.2023.10128325](https://doi.org/10.1109/IC-CCIS56745.2023.10128325).
- [81] M. M. U. Saleheen, M. S. Islam, R. Fahad, M. J. B. Belal, and R. Khan, "IoT-based smart agriculture monitoring system," in *Proc. IEEE Int. Conf. Artif. Intell. Eng. Technol.*, Kota Kinabalu, Malaysia, 2022, pp. 1–6, doi: [10.1109/IICAIET55139.2022.9936826](https://doi.org/10.1109/IICAIET55139.2022.9936826).
- [82] F. M. Shahrin, L. Zahin, R. Rahman, T. Roy, A. Khan, A. Khater, and M. D. Rahman, "IoT based smart automated agriculture and real time monitoring system," in *Proc. 2nd Int. Conf. Smart Electron. Commun.*, 2021, pp. 47–53, doi: [10.1109/ICOSEC51865.2021.9591855](https://doi.org/10.1109/ICOSEC51865.2021.9591855).
- [83] F. Shahrin, L. Zahin, R. Rahman, A. H. Kaf, and A. K. M. Abdul Malek Azad, "Agricultural analysis and crop yield prediction of Habiganj using multispectral bands of satellite imagery with machine learning," in *Proc. 11th Int. Conf. Elect. Comput. Eng.*, Dhaka, Bangladesh, 2020, pp. 21–24, doi: [10.1109/ICECE51571.2020.9393066](https://doi.org/10.1109/ICECE51571.2020.9393066).
- [84] T. T. Nguyen et al., "Monitoring agriculture areas with satellite images and deep learning," *Appl. Soft Comput.*, vol. 95, 2020, Art. no. 106565.
- [85] D. Murugan, A. Garg, and D. Singh, "Development of an adaptive approach for precision agriculture monitoring with drone and satellite data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5322–5328, Dec. 2017, doi: [10.1109/JSTARS.2017.2746185](https://doi.org/10.1109/JSTARS.2017.2746185).
- [86] Y. Karimi, S. O. Prasher, R. M. Patel, and S. H. Kim, "Application of support vector machine technology for weed and nitrogen stress detection in corn," *Comput. Electron. Agriculture*, vol. 51, no. 1/2, pp. 99–109, 2006.
- [87] P. B. Padol and A. A. Yadav, "SVM classifier based grape leaf disease detection," in *Proc. Conf. Adv. Signal Process.*, Pune, India, 2016, pp. 175–179, doi: [10.1109/CASP.2016.7746160](https://doi.org/10.1109/CASP.2016.7746160).
- [88] P. Radoglou-Grammatikis, P. Sarigiannidis, T. Lagkas, and I. Moscholios, "A compilation of UAV applications for precision agriculture," *Comput. Netw.*, vol. 172, 2020, Art. no. 107148, doi: [10.1016/j.comnet.2020.107148](https://doi.org/10.1016/j.comnet.2020.107148).
- [89] P. N. Patel, M. A. Patel, R. M. Faldu, and Y. R. Dave, "Quadcopter for agricultural surveillance," *Adv. Electron. Electr. Eng.*, vol. 3, pp. 427–432, 2013.
- [90] J. Huang, X. Wang, X. Li, H. Tian, and Z. Pan, "Remotely sensed rice yield prediction using multi-temporal NDVI data derived from NOAA's-AVHRR," *PLoS One*, vol. 8, 2013, Art. no. e70816, doi: [10.1371/journal.pone.007086](https://doi.org/10.1371/journal.pone.007086).
- [91] J. A. Paredes, J. Gonzalez, C. Saito, and A. Flores, "Multispectral imaging system with UAV integration capabilities for crop analysis," in *Proc. IEEE 1st Int. Symp. Geosci. Remote Sens.*, 2017, pp. 1–4, doi: [10.1109/GRSS-Chile.2017.7996009](https://doi.org/10.1109/GRSS-Chile.2017.7996009).
- [92] P. Radoglou-Grammatikis, P. Sarigiannidis, T. Lagkas, and I. Moscholios, "A compilation of UAV applications for precision agriculture," *Comput. Netw.*, vol. 172, 2020, Art. no. 107148, doi: [10.1016/j.comnet.2020.107148](https://doi.org/10.1016/j.comnet.2020.107148).
- [93] J. Su, M. Coombes, C. Liu, L. Guo, and W. H. Chen, "Wheat drought assessment by remote sensing imagery using unmanned aerial vehicle," in *Proc. Chin. Control Conf.*, 2018, pp. 10340–10344, doi: [10.23919/ChiCC.2018.8484005](https://doi.org/10.23919/ChiCC.2018.8484005).
- [94] O. Nevalainen et al., "Individual tree detection and classification with UAV-based photogrammetric point clouds and hyperspectral imaging," *Remote Sens.*, vol. 9, no. 3, 2017, Art. no. 185, doi: [10.3390/rs9030185](https://doi.org/10.3390/rs9030185).
- [95] V. S. Narwane, A. Gunasekaran, and B. B. Gardas, "Unlocking adoption challenges of IoT in Indian agricultural and food supply chain," *Smart Agricultural Technol.*, vol. 2, 2022, Art. no. 100035.
- [96] A. Agrawal, D. Arora, and P. Sharma, "Crop production prediction using machine learning: An Indian perspective," in *Proc. Emerg. Technol. Data Mining Inf. Secur.*, 2021, vol. 3, pp. 981–990.
- [97] K. Sharma, C. Sharma, S. Sharma, and E. Asenso, "Broadening the research pathways in smart agriculture: Predictive analysis using semiautomatic information modeling," *J. Sensors*, vol. 2022, 2022, Art. no. 5442865, doi: [10.1155/2022/5442865](https://doi.org/10.1155/2022/5442865).
- [98] J. Sasi Bhanu, S. D. Bigul, and A. Prakash, "Agricultural Internet of Things using machine learning," *AIP Conf. Proc.*, vol. 2358, no. 1, Jul. 2021, Art. no. 080010, doi: [10.1063/5.0058012](https://doi.org/10.1063/5.0058012).